

The Problem: AI Agents Need Memory

Every AI agent — chatbots, coding assistants, research tools, autonomous workflows — needs persistent memory. Today that means a vector database: Pinecone, Weaviate, Qdrant, or self-hosted FAISS. Each tenant requires gigabytes of storage, dedicated infrastructure, continuous index maintenance, and GPU compute for similarity search. As agentic AI scales to millions of users each needing their own memory, this becomes a hosting and cost problem with no current solution.

The Solution: Brain Cartridges

A brain cartridge is a single 12 MB file that is the memory. No server. No index build. No external dependencies. Mount it, search it, swap it. Each cartridge holds up to 1 million searchable patterns with built-in cryptographic integrity, per-pattern access controls, ownership metadata, and embedded source text.

Capability	Vector Databases	Brain Cartridges
Find similar items	Yes	Yes
Recall from partial/noisy input	No	Yes — attractor physics
Associative connections	No	Yes — automatic
Memory footprint	Grows linearly (GBs)	Fixed 12 MB per brain
Self-contained portable file	No (server + config)	Yes — one file
Per-item access control	Server-side only	Per-pattern, in-file
Tamper detection	No	SHA-256 manifest
GPU required	For similarity search at scale	No — CPU search modes available

Shipping Today — Four Open-Source Repos

Vector+ Studio — Full-stack search application (Python/React). Five search modes: Cosine+Hamming blend, physics-enhanced Smart Search, Pure Brain (GPU), Fast Cosine, and Associate (attractor neighborhood). Build cartridges from any text corpus. Mount, search, and manage brain cartridges through a polished UI.

Membot — MCP server giving AI agents physics-enhanced memory via swappable brain cartridges. Live on cloud infrastructure. Agents mount, search, store, and hot-swap domain-specific knowledge in milliseconds. Compatible with Claude, OpenClaw, and any MCP-enabled agent.

OpenClaw MCP Adapter — Plugin exposing MCP server tools as native agent tools, enabling seamless cartridge access from the OpenClaw agent framework.

Membrane — Defensive filter layer protecting AI agents browsing the web, with 5-layer defense against prompt injection and known attack vectors.

Proof It Works

1,000,000 real-world Wikipedia embeddings with perfect recall ($R@1 = 1.000$) under 10% noise. Validated across nine scale checkpoints from 2K to 1M with no degradation. Training-side noise immunity up to 50%. All quantization levels from float32 to 1-bit maintain perfect recall. Currently in use as persistent memory for our own AI development assistant (Claude Code), bridging context compactions with full recall.

Live demo examples: "Why did the Beatles break up?" returns Apple Records, John Lennon, The Beatles, Badfinger, Capitol Records, and Divorce — the lattice discovers conceptual associations that pure keyword and embedding search miss. Query "Is T. Rex a dinosaur or a band?" and results interleave both semantic domains by relevance.

Scale Economics

1,000-Tenant Deployment	Traditional (FAISS)	Brain Cartridges
Storage per tenant (1M patterns)	~4.8 GB	12 MB
GPUs required	63	1 (projected)
Annual cloud cost	\$828,000	\$13,100
CPU-only deployment	Limited (no GPU = no search)	Yes — Cosine+Hamming, no GPU

The CPU-only path is the near-term market entry for small to mid-size enterprises that cannot afford GPU infrastructure. Cosine+Hamming search on brain cartridges runs on any hardware, with the physics-enhanced modes available when GPU access is present. This scales down as cleanly as it scales up.

Go-to-Market

Near-term: Open-source engine with paid enterprise cartridge management. Two partnership conversations underway with Portland-area IT services firms interested in offering the cartridge suite to their lower-to-mid-tier client base. Active member of Portland AI Collective leadership.

Revenue model: Open-core. Free engine and tools. Paid tiers for enterprise cartridge management, multi-tenant hosting (Membot depot), priority support, and private cartridge registries. Future: marketplace commission on third-party cartridge sales.

Current Ask

\$25,000–\$50,000 angel funding to complete the production release, finalize benchmarks for publication, and convert partnership discussions into pilot deployments. Active engagement with Portland State Business Accelerator.

Next round: \$250K pre-seed to hire, expand enterprise pilots, and build the multi-tenant Membot depot for managed cartridge hosting.

Hardware roadmap: The same engine runs at every scale — from a \$12/month cloud droplet to purpose-built FPGA silicon. The 1-bit binary physics maps directly to Lattice Semiconductor Avant-series FPGAs with zero floating-point hardware. See companion document: "The 1-Bit Hardware Path."

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